RAMIDI SHIVA KARTHIK REDDY

Hyderabad

linkedin.com/in/shiva-karthik-undefined-50470a301

Education

Intermideate, *Sri gayatri Junior college 91%* Hyderabad, India

B.tech, *Guru Nanak Institute of Technology 2026* Hyderabad, India

Skills

PYTHON
Python: Anaconda, Jupyter Notebook, Colab Notebook, syntax, Identifiers & Operators, Array, Numpy, Data Manipulation, Pandas

SQL + Power BI
MYSQL, NoSQL, CRUD operation, RDBMS, Data Exploration and Data Filtering, DQL and Operators, Clauses, Joins, ACID, COMMIT, ROLLBACK.

CNN & Computer Vision
Open CV, Convolutional, TransferLearing, Object Detection, YOLO.

Advanced Statistics
Descriptive Statistics, Probability Distribution, Data Gathering Techniques, Inferential Statistics.

Machine Learning
Supervised, Unsupervised, ML Algorithms, Validation Methods, Naive Bayes, Linear Algebra, KNN, Hyperparameters, Dimensionality, Decision Tree, Linear regression, Evaluation Metrics, Regression Techniques

GEN AI (LLm's and Rag)
LLMs,Prompt engineering , Langchain , embeddings , vectorDB, Memory,Agents

Projects

EV Battery Failure Detection using Deep Learning

- Developed an **EV Battery Failure Prediction** model using **ANN, TensorFlow/Keras** on **200,000+ battery telemetry records**.
- Performed **EDA and data preprocessing** using **Pandas, NumPy, Matplotlib, and Seaborn**.
- Evaluated the model using **Precision, Recall, F1-Score, Confusion Matrix, and ROC-AUC**, achieving **97% accuracy and 0.99 ROC-AUC**.

Smart Waste Classification using CNN & YOLOv8

- Developed a **Smart Waste Classification System** using **CNN and MobileNetV2** to classify waste into **9 categories** from images.
- Performed **image preprocessing and data augmentation** using **TensorFlow/Keras** to improve model performance.
- Integrated **YOLOv8** for waste detection and CNN-based classification to enable automated waste segregation.
- Deployed the model using **Streamlit**, providing a user-friendly interface for real-time waste classification.

Certificates

- Exploratory Data Analysis from Innomatics Research Labs
- Certification in SQL Innomatics Research labs
- Machine Learning Certification from Innomatics research Labs